12. Bartlett Spectral Estimation

Simulate Bartlett spectral estimation for a signal divided into eight segments and evaluate variance reduction compared to the periodogram method. (25) (BL4) (WK4)

Given Data:

- **Signal Length = 4096 samples**
- **Sampling Frequency = 4 kHz**
- **Number of Segments = 8**

Tasks:

1. Divide the signal into equal segments.

2. Compute periodogram of each segment.

3. Average the spectra.

4. Plot Bartlett estimate.

5. Compare with standard periodogram.

6. Analyze variance reduction.

Aim

To implement Bartlett spectral estimation by dividing a signal into eight equal segments and compare its variance reduction with the standard periodogram.

Objective

1. To implement Bartlett spectral estimation.
2. To divide the signal into eight equal segments.
3. To compute the periodogram of each segment.
4. To average the spectra and obtain the Bartlett estimate.
5. To compare the Bartlett, estimate with the standard periodogram and analyze variance reduction.

Theory

Bartlett Spectral Estimation is a method used to estimate the **Power Spectral Density (PSD)** of a signal. Instead of using the entire signal at once, the signal is divided into several equal segments. The periodogram of each segment is calculated separately, and all the spectra are averaged to produce the final spectral estimate. This averaging reduces random fluctuations and variance, resulting in a smoother and more reliable spectrum than the standard periodogram. However, because each segment is shorter, the frequency resolution is slightly reduced.

Algorithm

1. Generate the input signal.
2. Set the sampling frequency to 4 kHz.
3. Divide the signal into 8 equal segments.
4. Compute the periodogram for each segment.
5. Average all the periodograms.
6. Plot the Bartlett spectral estimate.
7. Plot the standard periodogram.
8. Compare both spectral estimates.
9. Analyze the variance reduction.
10. Draw the conclusion.

Tool Used

Tool: MATLAB

About MATLAB

MATLAB (MATrix LABoratory) is a software developed by MathWorks for numerical computing and signal processing. It provides built-in functions to perform mathematical calculations, analyze signals, and display graphical results. MATLAB is widely used in engineering and research because it makes complex signal processing tasks simple and accurate.

Why MATLAB is Used

1. Easy to implement signal processing algorithms.
2. Provides built-in functions for spectral analysis.
3. Generates accurate graphs for comparison.
4. Supports FFT and periodogram calculations.
5. Easy to divide signals into segments.
6. Fast computation and visualization.
7. Widely used in DSP applications.

Functions Used :

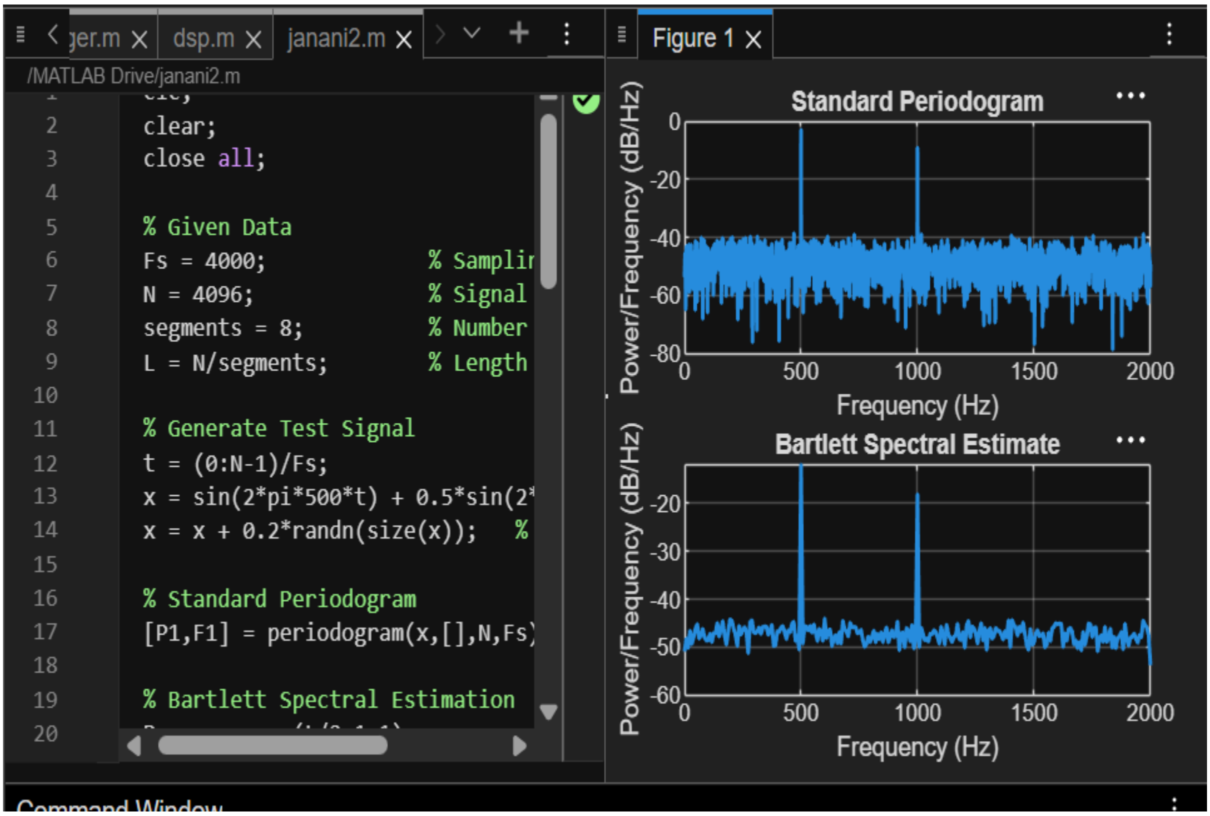
Function	Purpose
periodogram()	Computes the power spectral density of each segment
plot()	Displays the spectrum graph
subplot()	Shows multiple graphs in one figure
zeros()	Creates an array for averaging spectra
grid on	Adds grid lines to the graph

Code:

```
clc;
clear;
close all;
% Given Data
Fs = 4000;          % Sampling Frequency
N = 4096;           % Signal Length
segments = 8;       % Number of Segments
L = N/segments;     % Length of Each Segment
% Generate Test Signal
t = (0:N-1)/Fs;
x = sin(2*pi*500*t) + 0.5*sin(2*pi*1000*t);
x = x + 0.2*randn(size(x));   % Add Noise
% Standard Periodogram
[P1,F1] = periodogram(x,[],N,Fs);
% Bartlett Spectral Estimation
Pavg = zeros(L/2+1,1);
for k = 1:segments
    segment = x((k-1)*L+1 : k*L);
    [P,F] = periodogram(segment,[],L,Fs);
    Pavg = Pavg + P;
end
Pavg = Pavg/segments;
% Plot Results
figure;
subplot(2,1,1)
```

```
plot(F1,10*log10(P1),'LineWidth',1.5);

grid on;

title('Standard Periodogram');

xlabel('Frequency (Hz)');

ylabel('Power/Frequency (dB/Hz)');

subplot(2,1,2)

plot(F,10*log10(Pavg),'LineWidth',1.5);

grid on;

title('Bartlett Spectral Estimate');

xlabel('Frequency (Hz)');

ylabel('Power/Frequency (dB/Hz)');
```

output :



Result :

Bartlett spectral estimation was successfully implemented by dividing the signal into 8 equal segments. The averaged spectrum was smoother and showed lower variance than the standard periodogram. Although the frequency resolution was slightly reduced, the Bartlett method produced a more stable and reliable spectral estimate.